Cortical Decoders Trained on Other Individuals Age Slower Than Your Own

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Abstract

Cortical representations drift; a decoder fit to one animal's neural activity slowly loses accuracy on that animal over days, even with unchanged behavior. The standard solution is periodic recalibration on fresh data from the subject; we report a different form of robustness. On the open BraiDyn-BC cohort (25 mice, cortex-wide widefield $\Delta F/F$ parcellated into 44 Allen-atlas regions, a ∼15-day operant protocol), we decode four task events and compare, per mouse, how much accuracy is lost over two weeks by (i) the mouse's own early-week decoder and (ii) a decoder pooled over the other 24 mice's early-week data. The pooled decoder ages substantially less: there is a $+0.024$ AUC asymmetry between the personal and pooled models ($p = 2 \times 10^{-5}$; the own decoder ages quicker in 76 of 100 mouse–target pairs), and the effect holds for all four events ($p < 0.005$). A count-matched control tells us why: holding the number of training events fixed, a decoder from other mice ages far less than the subject's own. Trained on the mouse's own early data it loses 0.023 AUC over the two weeks; trained on the same amount of data from a single other mouse it loses none (-0.006). As both decoders see equal data, this is not a data-volume effect; it follows from not overfitting the subject's own drift-prone idiosyncrasies. The effect is not an artifact of a weak decoder: it holds, and stays significant, when the linear readout is replaced by a 1-D CNN or a GRU. Population pooling is thus a drift-mitigation strategy in its own right: the small early-week cost of decoding across individuals is repaid as temporal stability. Code and a streamed-feature pipeline are released.

1 Introduction

Representational drift, the reorganization of an animal's neural code over days with fixed behavior, is now documented across sensory, motor, and association cortex and in the hippocampus [14, 2, 16]. This has a stark impact on neural decoding: a decoder trained on an animal today degrades on that same animal weeks later. The dominant response is recalibration: periodically refitting the decoder on fresh, within-subject data [1, 4]. Recalibration requires ongoing labelled data from the individual, and it treats each subject in isolation.

We ask whether the code's cross-individual structure offers an alternative route to temporal robustness. Cortex-wide behavioral representations are known to be considerably shared among individuals [9, 6, 7], which allows models to decode behavior from unseen individuals [17]. The question we pose is temporal: does a decoder built from a population of other individuals resist drift better than a decoder built from the subject itself? If individual brains drift along idiosyncratic, partly independent trajectories, then averaging over many individuals should cancel the individual-specific component of drift. This prediction, to our knowledge, has not been examined.

We test it on BraiDyn-BC [5], a widefield calcium-imaging resource whose ∼15-day operant protocol and shared 44-region Allen parcellation make within-subject and cross-subject decoders directly comparable across time. Our contributions:

- **Pooling resists drift.** Across two weeks, a population-pooled decoder loses less accuracy than a mouse's own decoder, for all four decoded events (paired asymmetry $+0.024$ AUC, $p = 2 \times 10^{-5}$; Section 4.1), and the personal decoder's advantage erodes continuously across the 15-day protocol (Section 4.2).
- **It is not a data-volume artifact.** A count-matched control isolates the cause. At a fixed number of training events, a decoder trained on other mice ages far less than one trained on the subject's own (-0.006 vs. $+0.023$ AUC over two weeks). The gain therefore comes from training on individuals other than the subject, not from the larger training set pooling usually brings. Pooling over many mice rather than one lowers aging further (-0.012 vs. -0.006), a smaller, additional gain from a more diverse set of individuals (Sections 4.3–4.4).
- **It does not depend on the decoder.** The asymmetry stays positive and significant when the linear readout is replaced by a 1-D CNN or a GRU, and for two events it grows (Section 4.5).
- **A drift-mitigation framing.** The early-week accuracy loss of cross-subject decoding is repaid as late-week stability, suggesting population priors as an alternative to per-subject recalibration. We release code and a no-raw-download streaming pipeline.

2 Related work

Representational drift and its mitigation. Drift is characterized within individual subjects [14, 2, 16, 13], and most approaches to mitigating it also operate within the subject. These include manifold stabilization [1], supervised and self-calibrating recalibration [4, 12], and adversarial or nonlinear alignment to the subject's own reference day [3, 11]. Each method uses later data from the same subject to stabilize the decoder. We instead ask whether data from other subjects can provide that stability.

Cross-subject decoding. Shared structure across individuals already supports cross-subject readout through manifold alignment [8], contrastive embeddings [15], and, on this cohort, atlas-aligned foundation models [17]. Multi-participant pooling can also reduce overfitting: HTNet, for example, pools participant data to avoid fitting a decoder to a small single-subject sample [10]. Our question differs in two ways. First, our count-matched control keeps the training-set size fixed; the relevant overfitting is therefore not a consequence of limited data, but of relying on subject-specific features that later drift. Second, prior work measures accuracy or transfer at a fixed time. We measure how that accuracy changes over days, asking whether a pooled decoder ages more slowly.

Conservation of cortex-wide dynamics. Cortex-wide activity contains shared, low-dimensional, movement-related components [9, 6, 7]. Their presence makes cross-subject pooling possible, although it does not alone imply that pooling will resist drift.

3 Data and methods

Dataset. BraiDyn-BC [5] (DANDI:001425, CC-BY) provides, for each mouse and session, hemodynamically corrected dorsal-cortex $\Delta F/F$ traces parcellated into 44 Allen Common Coordinate Framework regions at ∼30 Hz, together with time-aligned behavioral channels. The dataset includes 25 mice, each of which completed ∼15 daily operant sessions over ∼3 weeks. We stream the 44-region traces and behavioral channels without downloading the raw movies.

Features and events. We detect the onset of four events (lever pull, lick, reward, and tone) as rising edges, with a 1 s refractory period. For each onset we construct a 44-dimensional evoked feature by subtracting the mean pre-onset baseline (-0.5–0 s) from the mean post-onset $\Delta F/F$ response (0–1 s) in each parcel. Negative samples are drawn from matched-count windows more than 2 s from any event. Every classification is a binary decode of one event against these matched negatives, scored by area under the ROC curve (AUC).

Table 1: Personal vs. pooled aging across days (AUC lost from early- to late-block test, holding training fixed), per event, $n = 25$ mice. Asymmetry = personal − pooled aging; a positive value means the mouse's own decoder ages more. p: paired sign-flip permutation.

Event	Personal aging	Pooled aging	Asymmetry (95% CI)	p	own ages more
Lever-pull	-0.012	-0.026	$+0.014$	0.002	68%
Lick	$+0.018$	-0.010	$+0.028$	0.0001	84%
Reward	$+0.023$	-0.010	$+0.033$	0.0001	80%
Tone	$+0.013$	-0.010	$+0.023$	0.005	72%
Pooled			$\mathbf{+0.024}$	$\mathbf{2 \times 10^{-5}}$	**76%**

Decoders. The primary decoder is a standardized ℓ_2-regularized logistic regression on the 44-dimensional evoked feature. This is a linear readout of the parcel-space code: it recovers the component of behavior that is linearly present in the population average, and, because it has no capacity to memorize spatiotemporal detail, it cannot inflate an apparent generalization difference through overfitting. It is also the natural comparison to prior cross-subject decoders, which are overwhelmingly linear [8, 10]. To test whether our conclusions survive a higher-capacity decoder, we additionally train two nonlinear models on the full evoked window (-0.5 to $+1.0\,\mathrm{s}$, 45×44): a 1-D convolutional network over time and a GRU over the frame sequence, each with a linear classification head (Section 4.5). All decoders are trained with the same events, blocks, and train/test splits, so the only variable is decoder capacity.

Temporal blocks and aging. We pool each mouse's sessions into an early block (days 1–5) and a late block (days 11–15), then report AUC under four conditions. Within-mouse same-block performance (WS) is measured by five-fold cross-validation on the early block. Within-mouse across-block performance (WX) is measured by training on the early block and testing on the late block. For pooled same-block performance (LS) we train on the early blocks of the other mice and test on the held-out mouse's early block; pooled across-block performance (LX) uses the same training data but tests on the held-out mouse's late block. We define aging as the loss in accuracy when testing moves from the held-out mouse's early block to its late block while training remains fixed. Personal aging is $\mathrm{WS} - \mathrm{WX}$, and pooled aging is $\mathrm{LS} - \mathrm{LX}$. For each mouse, the aging asymmetry is $(\mathrm{WS} - \mathrm{WX}) - (\mathrm{LS} - \mathrm{LX})$; a positive value means the personal decoder ages more. We assess significance with a paired sign-flip permutation test across mice (20,000 permutations) and compute 95% confidence intervals with a cluster bootstrap over mice.

4 Results

4.1 A population-pooled decoder ages more slowly than a personal one

For each mouse, we keep the test data fixed (first the mouse's early block, then its late block) and change only the source of the training data: either the mouse itself or the other 24 mice. The pooled decoder loses less accuracy across the two weeks (Table 1). The personal-minus-pooled aging asymmetry is positive for every event: $+0.014$ for lever pull, $+0.028$ for lick, $+0.033$ for reward, and $+0.023$ for tone. Each is significant at $p < 0.005$. Across all 100 mouse–target pairs, the asymmetry is $+0.024$ AUC ($p = 2 \times 10^{-5}$), with the personal decoder aging more in 76% of pairs. The same direction holds for lever pull, where both decoders improve with practice: the pooled decoder improves more. The personal decoder's early-week advantage therefore narrows by the late block.

4.2 The personal decoder's advantage erodes over the protocol

The block split compares two windows, but the effect should also appear as a continuous trend across all fifteen days. For each session day we compute the accuracy of the personal decoder and the pooled decoder and regress each on day; on early days the personal decoder is trained by leave-one-session-out, so it is never tested on a session it trained on. The personal decoder starts ahead: on day 1 it leads the pooled decoder by about 0.04 AUC, as expected for a decoder trained on the same animal. That lead then erodes. Across the protocol the personal decoder declines while the

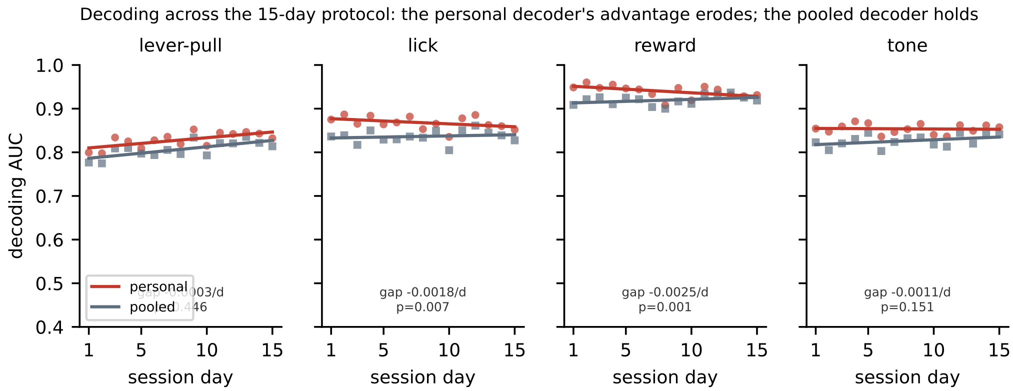

Figure 1: Decoding accuracy across the 15-day protocol for the personal decoder (own early data) and the pooled decoder (other mice's early data), with per-day points and linear fits. The personal decoder starts ahead and declines; the pooled decoder holds. Their gap narrows for lick and reward, where the personal decoder loses accuracy over days, and stays flat for lever pull and tone, where both decoders improve with practice. The per-event gap slope and its p (paired sign-flip over mice) are shown.

pooled decoder holds flat, and the gap between them narrows by 0.0014 AUC per day, pooled over all 100 mouse–target pairs ($p = 5 \times 10^{-5}$; the gap narrows in 64% of pairs). The trend is significant for lick and reward, where the personal decoder loses accuracy over days; for lever pull and tone both decoders improve with practice, so the gap stays flat (Fig. 1). By day 15 the personal decoder's early advantage has fallen to about 0.01 AUC.

4.3 The robustness is not a data-volume artifact

A pooled decoder contains $\sim 24\times$ more training events than a personal decoder. Its stability could therefore reflect data volume rather than pooling. We test this possibility with a count-matched control (Fig. 2, left). For each held-out mouse we set the training-set size to the number of events in that mouse's early block, n, and compare three decoders: self, trained on the mouse's own n events; one, trained on n events from one other mouse; and many, trained on n events drawn across the other 24 mice. All three are evaluated on the same early-to-late change in the held-out mouse. The ordering is consistent across all four events; averaged across targets, self ages by $+0.023$ AUC, one by -0.006, and many by -0.012. At the same sample count, the self decoder therefore ages $+0.029$ more than a decoder trained on one other mouse. Data volume cannot explain this gap. Instead, the mouse's early-week decoder appears to fit idiosyncratic features that later drift, while a decoder trained on another mouse does not have access to those features. Pooling across many mice yields a further -0.006 relative to one mouse (many $<$ one for all four events), so cross-individual diversity provides a smaller, additional benefit.

4.4 Diversity: a secondary scaling with the number of training individuals

We next vary the number of training mice, $N \in \{1, 2, 4, 8, 16, 24\}$ (Fig. 2, right). As N increases, pooled aging becomes slightly more favorable for lick (-0.001 to -0.010), reward (-0.004 to -0.010), and tone ($+0.003$ to -0.010). Lever-pull performance saturates with a single training mouse and remains near -0.025. This scaling effect is modest, approximately 0.01 AUC compared with the approximately 0.03 self-versus-other gap. Most of the resistance to drift is therefore present when training on only one other individual; additional individuals provide a smaller refinement.

4.5 The effect does not depend on the decoder

Every aging number so far comes from a linear decoder. A higher-capacity decoder could overfit the subject's early sessions more, which would enlarge the effect, or it could exploit fine temporal structure that averages out the effect. We recompute the personal-minus-pooled aging asymmetry

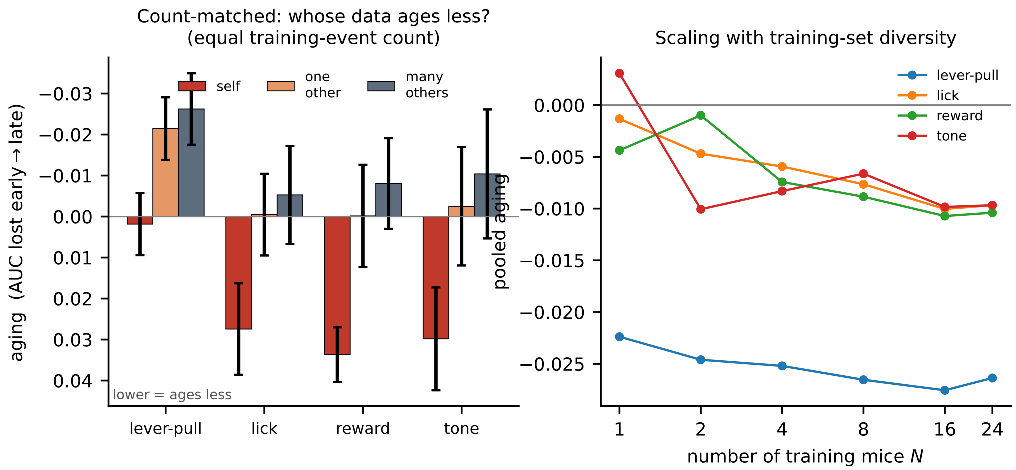

Figure 2: **Left:** count-matched control. With equal numbers of training events, decoder aging (AUC lost from early to late; lower values indicate greater robustness) is compared for the subject's own data (self), one other mouse, and many other mice. The self decoder ages, whereas the other-individual decoders do not. **Right:** pooled aging as a function of the number of training mice N. Greater diversity provides a small additional gain, with immediate saturation for lever pull.

Table 2: Personal-minus-pooled aging asymmetry under three decoders, per event, $n = 25$ mice. A positive value means the personal decoder ages more. Every cell is positive and significant; for lever pull and tone the effect is larger with the nonlinear decoders. p: paired sign-flip permutation.

Event	Linear	1-D CNN	GRU
Lever-pull	$+0.016$ (0.020)	$+0.088$ ($<10^{-4}$)	$+0.057$ ($<10^{-4}$)
Lick	$+0.033$ (0.0003)	$+0.028$ ($<10^{-4}$)	$+0.026$ ($<10^{-4}$)
Reward	$+0.031$ (0.0007)	$+0.028$ (0.0001)	$+0.019$ (0.002)
Tone	$+0.027$ (0.009)	$+0.061$ ($<10^{-4}$)	$+0.032$ (0.002)

with a 1-D CNN and a GRU over the full evoked window, on the same events and blocks (Table 2). The asymmetry stays positive and significant under every decoder and every event: the personal decoder ages more than the pooled one under the linear model, the CNN, and the GRU alike. For lever pull and tone the gap widens with the nonlinear decoders (CNN asymmetry $+0.088$ and $+0.061$), consistent with a higher-capacity decoder fitting the subject's drift-prone features more closely; for lick and reward it is comparable across decoders. The drift-resistance is therefore a property of the code, not of the linear readout.

5 Discussion

A subject's own decoder overfits idiosyncratic features of its early sessions, some of which later drift. A decoder that has never seen the subject must instead rely on components of the code that are shared across individuals and more stable over time. The count-matched control supports this reading directly: at equal data, other-individual data ages far less than the subject's own, and the effect survives a nonlinear decoder that has more capacity to overfit. Population structure is therefore not only a resource for cross-subject transfer, but also for temporal robustness. The initial accuracy cost of cross-subject decoding is recovered over time as the pooled decoder remains stable. Unlike within-subject recalibration, this approach does not require newly labelled data from the individual at test time.

Limitations. We measure the effect at the mesoscale, using parcel-averaged activity. Whether the same result holds for single-neuron codes, where representational drift is well characterized, remains unknown. The analysis also covers four discrete task events, a two-week interval, and one species; longer time periods, graded behaviors, other species, and other recording modalities should be tested. Our conclusions depend on within-versus-pooled and early-versus-late comparisons rather than on absolute AUC. Finally, although the count-matched control rules out training-set size and is consistent with self-overfitting, it does not identify which components of drift are individual-specific, and the additional effect of cross-individual diversity is small.

Reproducibility

BraiDyn-BC is publicly available through DANDI:001425. We release the analysis code and a streaming feature pipeline that reproduces all reported results from the parcellated traces without downloading the raw imaging data. Section 3 and the released code specify all data splits, permutation seeds, and hyperparameters.

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